

PAIND description

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- **Title:** Cold-water circulation with a heat pump
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- **Description:**

The project consists of a domestic water system with separate domestic cold water (DCW) and domestic hot water (DHW) distribution. The starting point of the project is the hygienic risk on the DCW side: during periods without flow, the DCW temperature can exceed the critical threshold of 20°C, creating conditions that may promote Legionella growth.

To address this, a closed DCW loop is introduced before the domestic distribution. The actively cooled DCW loop directly contributes to hygiene by keeping the cold-water temperature below the critical range. The heat extracted from the DCW loop can be reused for DHW preparation through a heat pump.

The DHW side includes a circulation loop to maintain hygiene and allow thermal disinfection against Legionella. The use of recovered heat for DHW production is a useful secondary benefit, but the main focus of the project is to assess whether active cooling of the DCW loop can improve hygienic safety while technically supporting heat pump operation.